4.1 SI base units

Table 1 gives the seven base quantities, assumed to be mutually independent, on which the SI is founded, and the names and symbols of their respective units, called "SI base units." Definitions of the SI base units are given in Appendix A. The kelvin and its symbol K are also used to express the value of a temperature interval or a temperature difference (see Sec. 8.5).

Table 1. SI base units

	SI base unit	
Base quantity	**Name**	**Symbol**
length	meter	m
mass	kilogram	kg
time	second	s
electric current	ampere	A
thermodynamic temperature	kelvin	K
amount of substance	mole	mol
luminous intensity	candela	cd

4.2 SI derived units

Derived units are expressed algebraically in terms of base units or other derived units. The symbols for derived units are obtained by means of the mathematical operations of multiplication and division. For example, the derived unit for the derived quantity molar mass (mass divided by amount of substance) is the kilogram per mole, symbol kg/mol. Additional examples of derived units expressed in terms of SI base units are given in Table 2. (The rules and style conventions for printing and using SI unit symbols are given in Secs. 6.1.1 to 6.1.8.)

Table 2. Examples of SI coherent derived units expressed in terms of SI base units

	SI coherent derived unit	
Derived quantity	**Name**	**Symbol**
area	square meter	m^2
volume	cubic meter	m^3
speed, velocity	meter per second	m/s
acceleration	meter per second squared	m/s^2
wavenumber	reciprocal meter	m^{-1}
density, mass density	kilogram per cubic meter	kg/m^3
specific volume	cubic meter per kilogram	m^3/kg
current density	ampere per square meter	A/m^2
magnetic field strength	ampere per meter	A/m
luminance	candela per square meter	cd/m^2
amount-of-substance concentration		
amount concentration , concentration	mole per cubic meter	mol/m^3

4.2.1 SI coherent derived units with special names and symbols

Certain SI coherent derived units have special names and symbols; these are given in Table 3. Consistent with the discussion in Sec. 4, the radian and steradian, which are the two former supplementary units, are included in Table 3. The last four units in Table 3 were introduced into the SI for reasons of safeguarding human health.